

Interpretation Meets Safety

How Interpretability Methods Understand and Enhance LLM Safety

Based on:

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*"Interpretation Meets Safety: A Survey on Interpretation Methods
and Tools for Improving LLM Safety"*

EMNLP 2025 — Nearly 70 works surveyed

My Role in Today's Presentation:

My teammates covered **what** interpretability is, and **what** safety risks and guardrails exist. I connect the two: how interpretability methods *actually* helps us build safer LLMs.

Why Bridge Interpretability and Safety?

The Problem with Existing Safety Methods

- RLHF, DPO, Constitutional AI work empirically — **but without understanding why**
- Models can be “deceptively aligned”: pass safety tests without internalizing safety (Hubinger et al.)
- Fine-tuning on just 100 harmful examples can erase all alignment
- We are patching symptoms, not fixing root causes

The Gap in Prior Surveys:

Surveys focus on *either* interpretability *or* safety, not how the former informs the latter. This is the first survey to bridge both systematically.

The Insight: Mechanistic Understanding = Better Fixes

Surgery, Not Chemotherapy

If we can **locate** the mechanism behind harmful behavior inside the model, we can **fix it precisely**:

- Find the “sycophancy” direction → steer it away
- Locate jailbreak-enabling attention heads → prune them
- Trace hallucination to training data → remove it
- Find bias SAE neurons → clamp them to zero

Survey covers nearly 70 works across four safety concerns:

Hallucination ∪ Jailbreaks & Harm ∪ Bias ∪ Privacy Leakage

Unified Framework: Interpret → Enhance → Tools



Four Safety Concerns Covered

- **Hallucination:** confident fabrication of false facts
- **Jailbreaks & Harmfulness:** bypassing safety alignment
- **Bias:** skewed or discriminatory outputs
- **Privacy Leakage:** memorized sensitive training data

Why these four share a root cause:

All stem from training data with risky patterns + objectives that favour memorization over generalization. Interpretation reveals *where* these manifest in model internals.

Interpretation Methods for Safety

How Do We Look Inside an LLM?

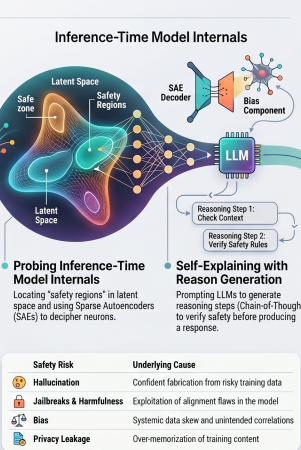
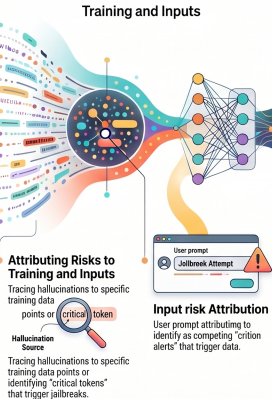
Training Attribution • Input Tokens • Model Internals • Self-Reasoning



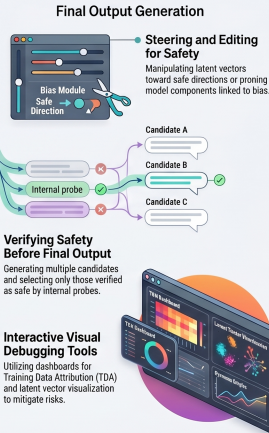
Interpretability for Safety: Overview

Interpretation Meets Safety: A Unified Framework for Secure LLMs

INTERPRETATION ACROSS THE LLM WORKFLOW



ENHANCING AND OPERATIONALIZING SAFETY



§3.1 Training Data Attribution (TDA)

Core hypothesis: unsafe outputs stem from problematic training data. Can we trace which examples caused them?

Three Attribution Approaches

Representation-Based

Compare similarity of training-example latent vectors to the output. Fast, but does not establish causality.

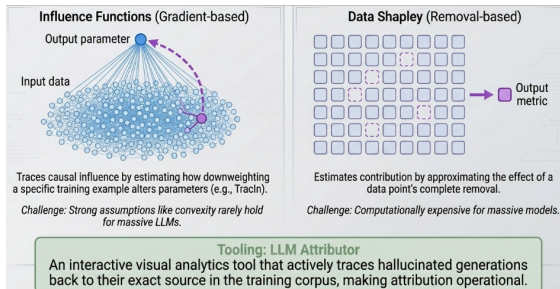
Gradient-Based

Influence Functions and TracIn estimate how removing a training example would affect model behavior.

Data Shapley

Measures each training example's contribution to model performance. Principled but computationally expensive.

Example Attribution Workflow



Once influential training examples are identified, they can be analyzed, filtered, or removed to improve model behavior.

Safety Applications

- **Hallucination Sources** LLM Attributor traces generated false facts back to specific training examples.
- **Bias Origins** Identifies documents introducing gender, racial, or cultural stereotypes.
- **Toxic Outputs** Links harmful generations to problematic training passages.
- **Training Dynamics** Tracks how capabilities such as refusal behavior emerge during training.

Open Challenge (§6.3):

Most frontier models use proprietary datasets. Using TDA to improve safety through data removal and retraining has only been demonstrated on relatively small models. Scaling remains unsolved due to the sheer magnitude of training data and the immense compute for each of these methods.

§3.2 Identifying Safety-Critical Input Tokens

Attribution Method Families

- **Attention Weights:** early and intuitive — higher weight $\approx$ more important. Aggregation (rollout, mean/max) reduces overhead across heads. *But: attention $\neq$ importance.*
- **Gradient-Based:** compute $\partial \text{output} / \partial \text{input embedding}$ — how much does each token shift the output? More faithful than raw attention.
- **Perturbation-Based:** mask/replace tokens and observe output changes. Model-agnostic, but creates out-of-distribution inputs.
- **Shapley Values:** fair attribution across token subsets. Extended to LLMs via TokenSHAP, TextGenSHAP.

Safety Applications

Jailbreak Token Detection

Cohen-Wang et al. (2024): perturbation-based attribution identifies the exact tokens triggering prompt poisoning attacks.

Wang et al. (2025): prompt LLMs to self-identify which input tokens activate jailbreak circuits — then remove them before generation.

Bias Diagnosis

Mohammadi (2024): Shapley values identify which tokens introduce gendered or racial bias — enabling targeted suppression without changing non-biased outputs.

Token attribution turns safety fixes into a **surgical filter**: remove only the offending tokens, preserve the rest of the user's intent.

§3.3.1 Probing Safety Regions in Latent Space

Probing Techniques

Mean Difference Vectors

Compute mean latent vector for harmful vs. harmless prompts. Their difference encodes the “harmfulness direction.”

Zou et al. (2023): Representation Engineering

Probing Classifiers

Train a linear classifier on latent vectors to predict safety properties. Successfully detects hallucinations, jailbreaks, and bias.

Known issue: poor cross-task generalization.

PCA / SVD on Activations

Dimensionality reduction uncovers the principal axes of unsafe behavior (Ball et al., 2024; Duan et al., 2024).

Key Safety Discoveries

The Refusal Direction (Arditi et al., 2024)

Jailbreak resistance is mediated by a **single linear direction** in activation space. Ablating it causes the model to comply with almost any harmful request.

Implication: robust safety = preserving this direction under adversarial pressure.

Universal Truthfulness Hyperplane (Liu et al., 2024)

A separating hyperplane exists in LLM latent space that distinguishes truthful from hallucinated generations — identifiable *before* the output is produced.

Enables real-time hallucination interception.

Probing is the **most widely used** interpretation method for safety — enabling real-time detection of unsafe generation in a single forward pass.

§3.3.2 Perturbing Components to Find Safety Mechanisms

Perturbation Methods

- **Component Knockout:** ablate layers, attention heads, or parameters to identify which are responsible for hallucinations, jailbreaks, and bias. **But what about polysemanticity?**
- **Activation Patching:** replace intermediate activations from one input with another input's activations, eg. safe vs unsafe flagged inputs, directly measures the causal effect of a specific region in terms of knowledge storage or biases induced.
- **Circuit Extraction:** build graphs of important components (nodes) and information flow (edges). Automates mechanistic analysis.
- **Parameter Perturbation:** modify weights; evaluate how safety properties change — tests alignment robustness.

Current Limitation:

Most circuit analysis targets simple grammar or arithmetic tasks. Very few studies address real-world safety problems at scale (Hanna et al., 2024).

Safety-Relevant Findings

Jailbreak-Critical Attention Heads

Zhao et al. (2024) and **Wei et al. (2024)**: specific attention heads mediate safety alignment. Pruning them enables jailbreaks; strengthening them improves robustness. Safety can be localized to specific heads per layer.

Bias Localization

Yang et al. (2023) and **Ma et al. (2023)**: component knockout pinpoints exact attention heads and MLP sublayers encoding gender and racial bias — enabling targeted debiasing without degrading other capabilities.

RAG Hallucination Circuits

Jin et al. (2024): activation patching localizes components responsible for ignoring retrieved context — explaining RAG faithfulness failures.

§3.3.3 Sparse Autoencoders: Deciphering What Neurons Encode

Most neurons are **polysemantic** — they encode multiple unrelated concepts simultaneously. SAEs disentangle them into interpretable features.

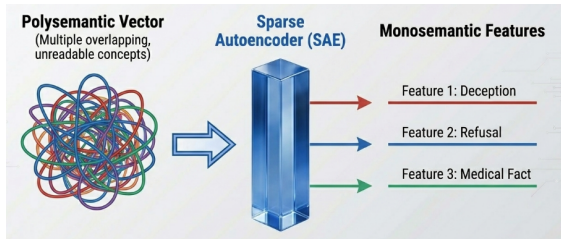
How SAEs Work

1. Encoder maps a polysemantic activation → sparse concept vector
2. Each dimension (SAE neuron) represents a distinct concept
3. Decoder reconstructs the original activation
4. Highly activating inputs reveal human-interpretable meanings

Anthropic's Scaling Monosemanticity (2024)

Applied to Claude 3 Sonnet, SAEs uncovered millions of interpretable features including deception, bias, sycophancy, dangerous content, and the famous *Golden Gate Bridge* feature.

Polysemantic → Monosemantic



§3.3.3 SAEs for Safety Intervention

Once interpretable features are identified, they can be manipulated directly to improve safety.

Jailbreak Features

Härle et al. (2024): SCAR identifies SAE features activated during jailbreak attempts. Suppressing them blocks attacks without retraining.

Bias Features

Hegde (2024) and **Zhou et al. (2025)** identify SAE features encoding gender bias. Clamping them reduces biased outputs.

Privacy Leakage Features

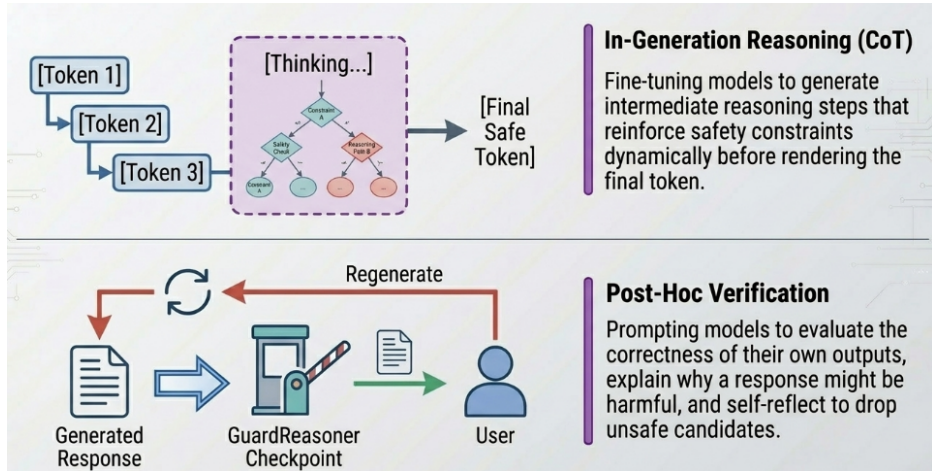
Frikha et al. (2025): PrivacyScalpel locates neurons encoding memorized personal information, enabling targeted privacy protection.

Also: Logit Lens

Projects intermediate hidden states onto the vocabulary, revealing what the model is “thinking” at each layer and helping detect hallucinations before generation completes.

§3.4 Self-Reasoning: LLMs Interpreting Their Own Outputs

LLMs can be prompted or trained to reason about their own outputs before or after generation.



Enhancing Safety via Interpretation

From Understanding to Action

Steer ∩ Modulate ∩ Edit ∩ Verify ∩ Self-Reason



§4.1 & §4.2.1 Token-Level Fixes and Representation Steering

§4.1 — Attend to Relevant Tokens

- Remove jailbreak-triggering tokens identified by attribution (*Pan et al., 2025*)
- Amplify attention to reliable user-specified tokens (*Zhang et al., 2024*)
- Redirect model focus to relevant context to reduce RAG hallucinations (*Liu et al., 2025*)

§4.2.1 — Representation Engineering

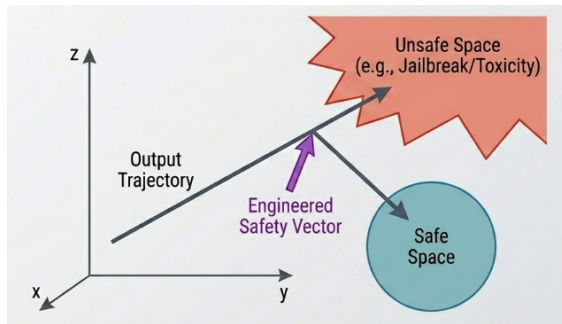
Steering Vectors

Add the “safe direction” discovered through probing directly to the hidden state:

$$h_{\text{new}} = h + \alpha \hat{v}_{\text{safe}}$$

No retraining required. Works entirely at inference time and can be tuned for different tasks.

Safety Vector Steering



§4.2.2 Modulating Neurons: SAE-Based Safety Control

The Mechanism

1. Run the SAE encoder to get a sparse concept vector for the current activation
2. Identify the SAE neuron(s) corresponding to the target concept (e.g., “deception”, “CSAM”, “violence”)
3. **Clamp, suppress, or amplify** those neurons in the sparse vector
4. Pass the modified concept vector through the SAE decoder to reconstruct a “safer” activation
5. Continue generation from this modified hidden state

No fine-tuning. No full retraining.
Targeted ð Reversible ð Interpretable.

Applications & Results

Refusal Steering — O’Brien et al., 2024

Identified the SAE neuron controlling **refusal behavior**. Amplifying it increases refusals on harmful requests. Suppressing it achieves near-universal compliance — a clear dual-use risk showing how powerful this control is.

Hallucination Mitigation — SAFE (Abdaljalil et al.)

SAE-based query enrichment suppresses hallucination-prone neurons in a RAG pipeline — improves factual accuracy without knowledge base changes.

Dual-Use Warning:

The same SAE control mechanisms that enhance safety can amplify harmful features or suppress safety features. Requires careful access controls and auditing.

§4.2 & §4.3 Verify Before Output + Self-Reasoning Safety

§4.2 — Verify Before Output

Best-of-N with Safety Filtering

Generate multiple candidate responses. Use probing classifiers or self-evaluation to select only safe ones.

SelfCheck (Miao et al., 2024): LLMs zero-shot check their own reasoning for internal inconsistency.

Chain-of-Verification (Dhuliawala et al., 2024): iterative fact-checking reduces hallucination substantially.

Intervention Mid-Generation

Detect unsafe token sequences using hidden-state probes *during* generation and **resample** from a safer distribution before committing to the harmful text.

§4.3 — Output with Self-Reasoning

GuardReasoner (Liu et al., 2025)

Fine-tune LLMs to generate intermediate reasoning explaining *why* a response may be harmful before deciding whether to comply. Makes safety decisions **transparent and auditable**.

CoT-Based Bias & Safety Reasoning

Kaneko et al. (2024): CoT prompting for gender bias evaluation — model articulates fairness reasoning explicitly.

Cao et al. (2024): CoT prompting for jailbreak defense — chain of thought acts as an in-context safety guardrail.

Self-reasoning safety methods produce **auditable** refusals: the decision comes with an explanation, not just a silent block.

Practical Tools

Operationalizing Safety-Focused Interpretation

Visualization ∩ Interaction ∩ Actionability



§5 Tools: Making Interpretation Actionable

§5.1 Training Data Attribution Visualizers

LLM Attributor (Lee et al., 2025): interactive tool tracing model outputs to training data. Users inspect which examples caused a hallucinated fact or toxic output — and can flag them for removal.

§5.2 Input Token Visualizations

Tools like **BertViz** (Vig, 2019), **iScore**, **PromptAID** (Mishra et al., 2025): visualize attention patterns across heads, reveal spurious token associations indicating bias. Support **interactive perturbation** — edit tokens, observe output shift live.

§5.3 Latent Vector Visualizations

Project latent vectors into 2D space; **logit lens** visualizers show vocabulary-level semantics at each layer; interactive steering tools let users push activations toward safer regions live. *LM-Debugger* (Geva et al., 2022); *Dashboard for Transparency* (Chen et al., 2024).

§5.4 Neuron & SAE Visualizations

Neuronpedia (Lin, 2023): interactive reference for SAE neuron concepts. **Neuroscope** (Nanda, 2022): browser for mechanistic interpretability of LLMs. Enable safety researchers to find, inspect, and manipulate safety-relevant features at scale.

Common thread: all categories enable **interactive** exploration — practitioners can act on findings, not just read about them.

Open Challenges & Future Directions

What Remains Unsolved?

Attacks • Evaluation • Training Attribution • Presentation • Safety Scope

6

Open Challenges at the InterpretabilitySafety Interface

[!] Interpretation-Based Attacks

Interpretation reveals *why* certain tokens trigger unsafe outputs — enabling adversaries to craft **more targeted** jailbreak suffixes (Lin et al., 2024; Winninger et al., 2025).

Adversaries use SAE feature maps to identify and manipulate model components linked to safety alignment (Arditi et al., 2024).

Tools built for transparency become attack vectors.

[?] Reliable Evaluation of Interpretations

No field consensus on “faithful” or “interpretable.” Evaluation scores vary with *presentation format*. Requires interdisciplinary collaboration: ML + HCI + cognitive science.

[DB] TDA for Safety Enhancement

TDA can trace unsafe behavior to training data — but **using that information to fix safety** (retrain without bad data) only demonstrated on small, non-generative models. Scaling to LLMs is an open problem.

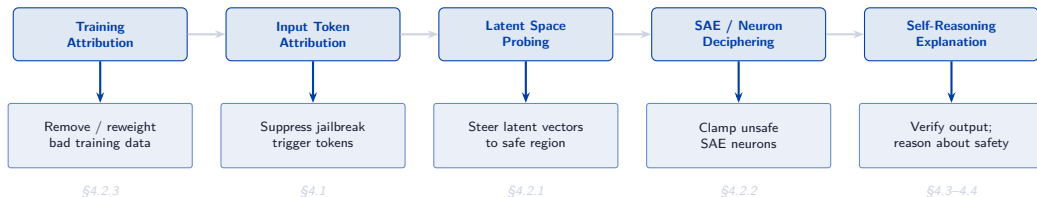
[People] User-Centered Presentation

Conversational interaction (e.g., TalkToModel) aids human understanding — but **no tools** apply this to safety-oriented interpretation. Different stakeholders (developers, auditors, regulators) need different views.

[Scope] Broader Safety Dimensions

Interpretation research focuses on hallucination, jailbreaks, bias, and privacy. **Under-studied:** code security, OOD robustness, and over-refusal (Siska & Sankaran, 2025).

Putting It All Together: The Full Interpret → Fix Pipeline



What Makes This Approach Powerful

- Mechanistic understanding, not just empirical patches
- Many methods work **at inference time** — no retraining
- Fixes are **targeted**: surgery, not chemotherapy
- Produces **auditable** safety decisions with explanations

What Are the Risks & Limits

- Dual-use: interpretation tools can enhance *or* undermine safety
- CoT explanations may not reflect actual model computation
- Scaling mechanistic circuit analysis to real tasks remains hard
- Non-linear safety representations challenge linear steering

Conclusion: Interpretation is the Path to Principled Safety

What We Covered

1. **§3 Interpretation Methods** (How we look inside)
TDA ð Token Attribution ð Probing ð Perturbation ð
SAEs ð Self-Reasoning
2. **§4 Safety Enhancements** (How we fix it)
Steering Vectors ð Neuron Modulation ð Model
Editing ð Output Verification
3. **§5 Practical Tools** (How we deploy it)
TDA Vis ð Token Vis ð Latent Vis ð Neuron/SAE
Explorers
4. **§6 Open Challenges**
Interpretation-based attacks ð Eval ð TDA scaling ð
Broader safety

The Core Message

Unsafe LLM behavior is not random, it has **identifiable, localizable causes** inside the model.

Interpretability gives us the tools to **find** those causes and **fix them precisely**.

Key Takeaways

- Interpretability $\neq$ only explaining models; it *enables* targeted safety fixes
- SAEs are becoming the workhorse of safety intervention, but carry dual-use risk
- Much unexplored: TDA at scale, circuit analysis for safety, holistic safety ontologies
- Interpretability and Safety research needs to go hand-in-hand

Reference: Interpretation Æ Safety Enhancement Summary

Interpretation Method	What It Reveals	How It Enhances Safety
Training Attribution	Poisonous training examples	Remove / reweight bad data (§4.2.3)
Input Token Attr.	Jailbreak trigger tokens	Suppress before generation (§4.1)
Latent Probing	Safety directions in activation space	Steer vectors; real-time detection (§4.2.1)
Component Perturbation	Harmful attention heads / layers	Prune / downscale components (§4.2.3)
SAEs	Polysemantic → monosemantic features	Clamp unsafe neurons precisely (§4.2.2)
Logit Lens	Layer-wise vocabulary predictions	Catch hallucinations mid-generation (§4.3)
Self-Reasoning (CoT)	Model's own safety assessment	Verify + explain before output (§4.4)

Source: Lee, Cho, Kim, Peng, Phute, Chau (Georgia Tech) ð *Interpretation Meets Safety* ð EMNLP 2025

Key References

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